

A Correlation-Production Reformulation of the Transfer-Entropy Second Law of Information Thermodynamics

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Abstract

Information thermodynamics extends the second law of thermodynamics to systems in which information processing plays a central role. Ito (2016) derived a generalized second law by employing transfer entropy and reversed transfer entropy as measures of information flow. While that derivation relies on the detailed fluctuation theorem, the present work provides an information-theoretic route to the same bound, under separability and non-synergistic bath-coupling assumptions, and explicitly accounts for correlations between the system and the heat bath following Esposito et al. (2010). Using an information-dynamical framework, the generalized second law of information thermodynamics is reformulated entirely in terms of entropy, mutual information, and information flow, thereby providing a transparent and intuitive interpretation of each contributing term.

Keywords: Information Thermodynamics, Transfer Entropy, Information dynamics, Information theory

1 Introduction

Information thermodynamics has extended the second law of thermodynamics to systems in which information processing plays an essential role [1]. Foundational studies on Maxwell’s demon [2], Szilard’s engine [3], and Landauer’s principle [4, 5] clarified the thermodynamic implications of measurement and erasure processes. These developments established that apparent violations of the second law may arise

when information processing is explicitly taken into account [6–13].

By characterizing information processing in terms of information flow, previous studies formulated generalized second laws using the measure for causality, the transfer entropy [14, 15]. In particular, Ito and Sagawa [16] and Hartich et al. [17] derived inequalities that bound entropy production by transfer entropy. Subsequently, Ito [18] demonstrated that a tighter bound can be obtained by incorporating reversed transfer entropy (also referred to as backward transfer entropy) [19–21]. These results rely crucially on the detailed fluctuation theorem [22–24], which relates entropy production to the probability ratio of forward and backward trajectories. Despite its formal exactness, the fluctuation-theorem-based formulation presents conceptual and practical challenges. Since it is expressed in terms of backward path probabilities, it obscures a purely information-theoretic interpretation of entropy production. Such difficulties naturally arise in strongly coupled systems and in quantum thermodynamics, where the definition of time-reversed dynamics is not unique.

An alternative perspective has been developed within quantum information theory, where entropy production is interpreted in terms of correlation production between a system and its environment [25–29]. Under deterministic and reversible total dynamics governed by Liouville’s theorem [30, 31], these studies have shown that entropy production can be identified with the production of mutual information, provided that the heat bath is sufficiently large.

Although Ref. [18] and the correlation-production approach both describe entropy production with information-theoretic quantities, their direct relationship has not been made explicit. The present work fills this gap by deriving the generalized second law derived in Ref. [18] from information-dynamical identities and explicit system-bath correlations. The derivation is based on an information-dynamical framework developed in Ref. [32], which provides exact evolution equations for entropy, mutual information, transfer entropy, and reversed transfer entropy. This framework enables a systematic and fully information-theoretic derivation of the generalized second law without explicit reliance on the detailed fluctuation theorem, while explicitly revealing how dynamical assumptions, such as separability, determinism, and system-bath correlations, enter the formulation.

2 Definitions

This paper follows the definitions and notation of Ref. [32] for the information-theoretic quantities used in the subsequent derivation. For completeness,

$$\begin{aligned}
\text{Individual Entropy [33, 34] :} & \quad H(X) := - \sum_{x \in M} P(X = x) \ln P(X = x) \\
\text{Conditional Entropy [33, 34] :} & \quad H(X|Y) := H(X, Y) - H(Y) \\
\text{Mutual Information [34] :} & \quad I(X, Y) := H(X) + H(Y) - H(X, Y) \\
\text{Conditional Mutual Information [34] :} & \quad I(X, Y|Z) := H(X|Z) + H(Y|Z) - H(X, Y|Z) \\
\text{Transfer Entropy [14, 15] :} & \quad T_{X \rightarrow Y} := I(X, Y'|Y) \\
\text{Reversed Transfer Entropy [18–20] :} & \quad T_{X \rightarrow Y}^\dagger := I(X', Y|Y') \\
\text{Conditional Transfer Entropy :} & \quad T_{X \rightarrow Y|Z} := I(X, Y'|YZ) \\
\text{Conditional Reversed Transfer Entropy :} & \quad T_{X \rightarrow Y|Z}^\dagger := I(X', Y|Y'Z),
\end{aligned} \tag{1}$$

where M is a discrete set, indicating that the above quantities are non-negative, and primed variables are evaluated one time step after the evaluation of unprimed variables.

While the above mutual information and transfer entropy represent the pair-wise correlation and causality, the complete information dynamics of a many-body system requires treatment of multivariate correlations and the relevant causal structure, which are responsible for multivariate effects, such as synergy or redundancy. As an information-theoretic quantity for multi-correlations, the studies [35–42] have investigated the co-information (interaction information or multiple mutual information). While the research about partial information decomposition (PID) has suggested other non-negative information-theoretic quantities with a comprehensive interpretation, the present work adopts the multiple mutual information, since the quantities of PID are not unique.

For the subsequent definitions, the entropy of a set of random variables is defined using the joint probability distribution: $H(V) := - \sum_{v_1, \dots, v_m \in M} P((V_1, \dots, V_m) = (v_1, \dots, v_m)) \ln P((V_1, \dots, V_m) = (v_1, \dots, v_m))$. Then,

$$\begin{aligned}
\text{Co-information [35–42] :} & \quad I(V) := \sum_{i=1}^{|V|} (-1)^{i+1} \sum_{U \subset V, |U|=i} H(U) \\
\text{Conditional co-information :} & \quad I(V|W) := \sum_{i=1}^{|V|} (-1)^{i+1} \sum_{U \subset V, |U|=i} H(U|W) \\
\text{Multiple Transfer Entropy [32] :} & \quad T_{V \rightarrow U} := I(V \cup U'|U) \\
\text{Multiple Reversed Transfer Entropy [32] :} & \quad T_{V \rightarrow U}^\dagger := I(V' \cup U|U'),
\end{aligned} \tag{2}$$

where $U' := \{u' | u \in U\}$.

3 Derivation of the Generalized Second Law

3.1 Problem Setting

We consider two systems X and Y , and a heat bath B coupled to system X . At the initial time $t = t_0$, the system and bath are assumed to be uncorrelated, such that

$$\begin{aligned} I(X, B) &= 0 \\ I(Y, B) &= 0 \\ I(Z, B) &= 0 \text{ with } Z = (X, Y) \\ \therefore I(Y, B|X) &= 0. \end{aligned} \tag{3}$$

Following Ref. [18], the dynamics of systems X and Y is assumed to satisfy the separability condition:

$$P(\Gamma'_X, \Gamma'_Y | \Gamma_X, \Gamma_Y) = P(\Gamma'_X | \Gamma_X, \Gamma_Y) P(\Gamma'_Y | \Gamma_X, \Gamma_Y). \tag{4}$$

Since the bath interacts only with system X , it is assumed to be conditionally independent of system Y given the states of X ,

$$\begin{aligned} T_{B \rightarrow Y|XX'} &= I(B, Y' | Y, X, X') = 0 \\ T_{B \rightarrow Y|XX'}^\dagger &= I(B', Y | Y', X, X') = 0. \end{aligned} \tag{5}$$

3.2 Information Dynamic Equations

Several exact identities are required for the derivation of the generalized second law. These identities form the equations of the information-dynamical framework introduced in Ref. [32].

Identity 1.

$$\Delta I(X, Y) = [T_{X \rightarrow Y} - T_{X \rightarrow Y}^\dagger] + [T_{Y \rightarrow X} - T_{Y \rightarrow X}^\dagger] + \alpha_2^{X, Y}, \tag{6}$$

where $\Delta I(X, Y) := I(X', Y') - I(X, Y)$ and

$$\alpha_2^{X, Y} := I(X', Y' | X, Y) - I(X, Y | X', Y'). \tag{7}$$

(Proof) Section III. A in the study [32]. ■

In the presence of three subsystems X , Y , and B , multi-body correlations naturally arise. To treat these correlations, conditional mutual information and conditional transfer entropy are employed, rather than multiple mutual information and multiple transfer entropy [32].

Identity 2.

$$\Delta H(X, B) = [T_{Y \rightarrow X} - T_{Y \rightarrow X}^\dagger] + I(Y', B'|X') - \alpha_2^{Y, B|X, X'} + \alpha_1^{\{Y\} \rightarrow X, B}, \quad (8)$$

where $\Delta H(X, B) := H(X', B') - H(X, B)$ and

$$\begin{aligned} \alpha_2^{Y, B|X, X'} &:= I(Y', B'|Y, B, X, X') - I(Y, B|Y', B', X, X') \\ \alpha_1^{\{Y\} \rightarrow X, B} &:= H(X', B'|X, B, Y) - H(X, B|X', B', Y'). \end{aligned} \quad (9)$$

(Proof)

$$\begin{aligned} \Delta H(X, B) &= [T_{Y \rightarrow X, B} - T_{Y \rightarrow X, B}^\dagger] + \alpha_1^{\{Y\} \rightarrow X, B} \\ &= [T_{Y \rightarrow X|B} + T_{Y \rightarrow B|X, X'}] - [T_{Y \rightarrow X|B'}^\dagger + T_{Y \rightarrow B|X, X'}^\dagger] + \alpha_1^{\{Y\} \rightarrow X, B}. \end{aligned} \quad (10)$$

The conditional transfer entropy terms $T_{Y \rightarrow X|B}$ and $T_{Y \rightarrow X|B'}^\dagger$ can be written as:

$$\begin{aligned} T_{Y \rightarrow X|B} &= T_{Y \rightarrow X} - [I(Y, B|X) - I(Y, B|X, X')] \\ &= T_{Y \rightarrow X} + I(Y, B|X, X') \quad (\because \text{Eq.(3)}) \\ T_{Y \rightarrow X|B'}^\dagger &= T_{Y \rightarrow X}^\dagger - [I(Y', B'|X') - I(Y', B'|X, X')] \end{aligned} \quad (11)$$

Using an identity analogous to Eq. (6), one obtains

$$I(Y', B'|X, X') - I(Y, B|X, X') = [T_{Y \rightarrow B|X, X'} - T_{Y \rightarrow B|X, X'}^\dagger] + [T_{B \rightarrow Y|X, X'} - T_{B \rightarrow Y|X, X'}^\dagger] + \alpha_2^{Y, B|X, X'}. \quad (12)$$

Substituting these relations yields

$$\begin{aligned} \Delta H(X, B) &= T_{Y \rightarrow X} - T_{Y \rightarrow X}^\dagger + I(Y', B'|X') - [T_{B \rightarrow Y|X, X'} - T_{B \rightarrow Y|X, X'}^\dagger] - \alpha_2^{Y, B|X, X'} + \alpha_1^{\{Y\} \rightarrow X, B} \\ &= T_{Y \rightarrow X} - T_{Y \rightarrow X}^\dagger + I(Y', B'|X') - \alpha_2^{Y, B|X, X'} + \alpha_1^{\{Y\} \rightarrow X, B} \quad (\because \text{Eq.(5)}) \blacksquare \end{aligned} \quad (13)$$

3.3 The Generalized Second Law of Information Thermodynamics

Since $I(X, B) = 0$, the entropy change satisfies $\Delta H(X) + \Delta H(B) = \Delta H(X, B) + I(X', B')$. Using Identity 2, the quantity $-\Delta H(X) - \Delta H(B)$ can be expressed as:

$$\begin{aligned} -\Delta H(X) - \Delta H(B) &= -[T_{Y \rightarrow X} - T_{Y \rightarrow X}^\dagger] - I(Y', B'|X') + \alpha_2^{Y, B|X, X'} - \alpha_1^{\{Y\} \rightarrow X, B} - I(X', B') \\ &= [T_{X \rightarrow Y} - T_{X \rightarrow Y}^\dagger] - \Delta I(X, Y) + \alpha_2^{X, Y} - I(Y', B'|X') + \alpha_2^{Y, B|X, X'} - \alpha_1^{\{Y\} \rightarrow X, B} - I(X', B') \end{aligned} \quad (14)$$

Here, the terms $\alpha_2^{X, Y}$, $I(Y', B'|X')$, $I(X', B')$, $\alpha_2^{Y, B|X, X'}$, and $\alpha_1^{\{Y\} \rightarrow X, B}$ are evaluated.

As the separability condition, Eq. (4), implies $I(X', Y'|X, Y) = 0$,

$$\alpha_2^{X,Y} = -I(X, Y|X', Y') \leq 0. \quad (15)$$

And by non-negativity of mutual information,

$$\begin{aligned} I(Y', B'|X') &\geq 0. \\ I(X', B') &\geq 0. \end{aligned} \quad (16)$$

The sum $I(X', B') + I(Y', B'|X') = I(Z', B')$, where $Z' = (X', Y')$, corresponds to the mutual information production discussed in quantum information theory [25–29]. Collecting terms yields

$$-\Delta H(X) - \Delta H(B) \leq [T_{X \rightarrow Y} - T_{X \rightarrow Y}^\dagger] - \Delta I(X, Y) + \alpha_2^{Y,B|X,X'} - \alpha_1^{\{Y\} \rightarrow X,B} \quad (17)$$

Up to this point, no thermodynamic assumptions concerning the heat bath have been imposed, and Eq. (17) follows solely from the information-dynamical setting introduced in Sec. 3.1. In particular, the quantity $\Delta H(X) + \Delta H(B)$ appearing in Eq. (17) does not yet correspond to the thermodynamic entropy production.

In Ref. [18], this quantity is evaluated using the detailed fluctuation theorem, yielding

$$\sigma_{DFT} = \Delta H(X) + \delta, \quad (18)$$

with

$$\delta = \sum_{x,y,x'} P(X' = x', X = x, Y = y) \ln \frac{P(X' = x'|X = x, Y = y)}{P(X' = x|X = x', Y = y)}, \quad (19)$$

the term δ cannot be expressed solely in terms of entropy, mutual information, transfer entropy, or reversed transfer entropy. Here, instead, the entropy change of the bath is evaluated following the information-theoretic approach of Ref. [25]. The bath is assumed to be initially in thermal equilibrium, described by the canonical distribution

$$P(B = \Gamma_B) = \frac{1}{Z} e^{-\beta E_B(\Gamma_B)}, \quad (20)$$

where β is the inverse temperature and $E_B(\Gamma_B)$ is the energy of the bath and Z is the partition function.

The entropy change of the bath is then given by

$$\begin{aligned} \Delta H(B) &= H(B') - H(B) \\ &= -\beta Q - D(B'|B), \end{aligned} \quad (21)$$

where

$$-\beta Q := \sum_{\Gamma_B} [P(B = \Gamma_B) - P(B' = \Gamma_B)] \ln P(B = \Gamma_B), \quad (22)$$

and $D(B'||B)$ denotes the relative entropy:

$$D(B'||B) := \sum_{\Gamma_B} P(B' = \Gamma_B) \ln \frac{P(B' = \Gamma_B)}{P(B = \Gamma_B)}. \quad (23)$$

Combining Eqs. (20) and (22) yields

$$Q = \left[\sum_{\Gamma_B} P(B = \Gamma_B) E_B(\Gamma_B) - \sum_{\Gamma'_B} P(B' = \Gamma'_B) E_B(\Gamma'_B) \right], \quad (24)$$

which identifies Q as the heat exchanged from the bath to the system.

For a sufficiently large heat bath with the size N_B , the relative entropy term $D(B'||B)$ becomes negligible [27, 43, 44]. To demonstrate this, assume that the bath distribution deviates slightly from equilibrium as:

$$P(B' = \Gamma_B) = P(B = \Gamma_B) + \epsilon \zeta(\Gamma_B), \quad (25)$$

where $\epsilon \propto 1/N_B$ and $\zeta(\Gamma_B)$ is an arbitrary function such that $\sum_{\Gamma_B} \zeta(\Gamma_B) = 0$. Then,

$$\beta Q = -\epsilon \beta \sum_{\Gamma_B} \zeta(\Gamma_B) E_B(\Gamma_B), \quad (26)$$

while the relative entropy scales as,

$$D(B'||B) \sim \epsilon^2 \sum_{\Gamma_B} \frac{\zeta^2(\Gamma_B)}{P(B = \Gamma_B)} + O(\epsilon^3) = O(\epsilon^2). \quad (27)$$

Under this scaling, the relative-entropy correction is second order in the bath perturbation, whereas the heat term is first order. Therefore, in the large-bath limit considered here, $D(B'||B)$ is neglected and $\Delta H(B) \sim -\beta Q$.

The conventional thermodynamic entropy production σ is defined as [45, 46]

$$\sigma = \Delta H(X) - \beta Q. \quad (28)$$

Substituting the expression for $\Delta H(B)$ into Eq. (17) yields

$$-\sigma \leq [T_{X \rightarrow Y} - T_{X \rightarrow Y}^\dagger] - \Delta I(X, Y) + \alpha_2^{Y, B|X, X'} - \alpha_1^{\{Y\} \rightarrow X, B}. \quad (29)$$

When the total dynamics is deterministic and reversible, as ensured by Liouville's equation, the following conditional entropies vanish [32]:

$$H(B', X', Y'|B, X, Y) = H(B, X, Y|B', X', Y') = 0. \quad (30)$$

Since $H(B', X'|B, X, Y) \leq H(B', X', Y'|B, X, Y)$ and $H(B, X|B', X', Y') \leq H(B, X, Y|B', X', Y')$, one obtains:

$$\alpha_1^{\{Y\} \rightarrow X, B} = H(B', X'|B, X, Y) - H(B, X|B', X', Y') = 0. \quad (31)$$

Analogously, the following quantities vanish:

$$I(Y', B'|Y, B, X, X') = I(Y, B|Y', B', X, X') = 0. \quad (32)$$

Therefore,

$$\alpha_2^{Y, B|X, X'} = 0. \quad (33)$$

As a result, under the above assumptions,

$$-\sigma \leq [T_{X \rightarrow Y} - T_{X \rightarrow Y}^\dagger] - \Delta I(X, Y). \quad (34)$$

The resulting inequality coincides with that derived in Ref. [18] for zero time delay between systems X and Y and two discrete time steps. To establish the equivalence $\sigma = \sigma_{DFT}$, consider a classical thermal bath satisfying

$$\frac{P(X' = x'|X = x, Y = y)}{P(X' = x|X = x', Y = y)} = e^{-\beta(E_X(x') - E_X(x))}. \quad (35)$$

With conservation of total energy, $E_{total} = E_B + E_X + V_{BX}$ where V_{BX} is the interaction energy, the difference between the two entropy productions is given by

$$\sigma - \sigma_{DFT} = \sum_{\Gamma_B, x} P(B = \Gamma_B, X = x) V_{BX}(\Gamma_B, x) - \sum_{\Gamma'_B, x'} P(B' = \Gamma'_B, X' = x') V_{BX}(\Gamma'_B, x'). \quad (36)$$

Therefore, $\sigma = \sigma_{DFT}$ holds in the weak-coupling or high-temperature limit. Beyond this regime, the equivalence generally fails. In quantum thermodynamics, σ_{DFT} further depends on the definition of backward trajectories [28], whereas the present approach remains well-defined. Similarly, in stochastic dynamics, σ_{DFT} is readily computed, while evaluating σ requires explicit modeling of bath dynamics.

The present framework may be advantageous in situations involving strong system-bath coupling or ambiguity in the definition of backward paths. Although it requires explicit information about system-bath correlations, this feature enables the evaluation of entropy production in systems driven by non-thermal or structured noise sources. Moreover, the information-dynamical formulation makes transparent how dynamical assumptions, such as separability, determinism, and reversibility, enter the generalized second law.

4 Conclusion

In this work, the generalized second law originally derived by Ito (2016) is reformulated within an information-theoretic framework that explicitly relates entropy production to correlation production. Rather than relying on the detailed fluctuation theorem, the derivation is based on exact information-dynamical identities and an explicit treatment of system-bath correlations. Although this approach assumes deterministic and reversible total dynamics in accordance with Liouville's theorem, it remains formally well defined in regimes where strong system-bath coupling or

ambiguities in the definition of backward paths render fluctuation-theorem-based formulations less transparent, as in quantum thermodynamics.

The results, Eqs. (14) and (34), do not yet constitute a closed information-dynamical description, since the mutual information between the system and the bath at the final time and the associated α -terms are not determined solely by information-theoretic quantities, such as H, I, T , and $T^\dagger$, at the initial time.

A natural direction for future work is to apply this framework to concrete models of Maxwell’s demon and Landauer erasure [47], thereby elucidating the precise relationship between information processing operations and information flow quantified by transfer entropy and reversed transfer entropy.

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